

```

e) int Function1 (int b[], int size)
{
    if (size == 0)
        return b[0];
    else
        return b[size-1];
}

void Function2 (int b[], int current, int s)
{
    for (int i=0; i<s; i++) // s=5
    {
        int n = Function1 (b, i);
        cout << b[current] << " " << n << endl;
    }
}

int main()
{
    const int Size=5;
    int a[Size]={1, 2, 3, 4, 5};
    cout << "Array contents are : ";
    Function2 (a, 0, Size);
    cout << endl;
    int result = Function1 (a, Size);
    cout << "Result is " << result;
    cout << endl;
    return 0;
}

```

$i = 0$   
 $\{1, 2, 3, 4, 5\}$   
 if (0);  
 return  
 1;  
 $b[4] = 5$

$n = 5$   
 $b[4] = 5$   
 bE

①  $b[0]$  — 1  
 ②  $b[0]$  — 1  
 ③  $b[1]$  — 2  
 ④  $b[2]$  — 3  
 ⑤  $b[3]$  — 4  
 [Marks: 15]

Q. # 2:

Write a function named "location\_of\_target" that takes as its arguments the following:

- (1) an array of integer values;
- (2) an integer that tells how many integer values are in the array;
- (3) an integer "target value".

The function should determine whether the given target value occurs in any of the cells of the array, and if it does, the function should return the subscript of the cell containing the target value. If more than one of the cells contains the target value, then the function should return the largest subscript of the cells that contain the target value. If the target value does not occur in any of the cells, then the function should return the sentinel value -1. Thus, for example, if the target value that's passed to the function is 34 and the array that's passed to the function looks like this:



**Final Exam: EE 116 Introduction to Computing  
: EE 110 Programming for Engineers - I**

Time: 3 hours, Max. Marks: 100, Spring 2016

**Instructions:**

- This question paper has 5 questions.
- Number your answers correctly and clearly on the answer sheet.
- Answer all parts of a question together.
- Use black or blue pen to answer the questions.

[Marks: 4+4+4+6+7=25]

**Q. # 1:**

What is the output of the following code segments? Show your work.

a)

```
int n, k = 5;  
n = (100 % k ? k + 1 : k - 1);  
cout << "n = " << n << " k = " << k << endl;
```

b)

```
int i = 5, j = 6, k = 7, n = 3;  
cout << i + j * k - k % n << endl;  
cout << i / n << endl;
```

c)

```
int d = 28;  
while (d%2==0 || d > 1)  
{  
    d = d / 2;  
    cout << d << "\n";  
}
```

d)

```
int Mystery (const char s[])  
{  
    int x=0;  
    while (s[x]!='\0')  
        x++;  
    return x;  
}  
  
int main()  
{  
    char str[80]="Programming";  
    cout << Mystery (str);  
}
```

*int 80  
char 80*

|    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|
| 0  | 1  | 2  | 3  | 4  | 5  | 6  |
| 58 | 26 | 91 | 34 | 70 | 34 | 88 |

then the target value occurs in cells 3 and 5, so the function should return the integer value 5.

**Q. # 3:**

[Marks: 15]

Write a function named "sort3" that takes three floating point arguments, call them "x", "y", and "z", and modifies their values, if necessary, in such a way as to make true the following inequalities:  $x < y < z$ .

The function should return no value. To take an example, if the following code fragment is executed

```
float a = 3.2, b = 5.8, c = 0.9;
```

```
sort3 (a, b, c);
```

```
cout << a << " " << b << " " << c << endl;
```

then the output will be

```
0.9 3.2 5.8
```

The function sort3 is allowed to make calls to other functions that you have written.

**Q. # 4:**

[Marks: 10+ 15=25]

- Write the algorithmic steps (in english) for a program that reads a positive integer entered by a user and then prints out all the positive divisors of that integer in a column in decreasing order.
- Implement the above algorithm in C++ (You should use functions for this part otherwise half of the marks would be deducted). Here is a sample output to demonstrate how the program works

Please enter a positive integer: 36

```
36
18
12
9
6
4
3
2
1
```

Q. # 5:

[Marks: 20]

Write a function named "number\_of\_matches" that compares the initial parts of two cstrings to see how many pairs of cells match before a difference occurs. For example, if the cstrings are

gain gains

then the function should return the value 4 because only the first four characters in the cstrings match. The pairwise cell comparisons should not go beyond the end of either cstring. The function should take only two parameters, namely the two cstrings to be compared (do not use the string library).

-----END-----

```
if for (int i = 1; i  
while (i != '\0')  
{
```

```
char a;  
char a[10];  
char b[10];
```

```
char a;  
a = 'gain';  
char b;  
b = 'gains';
```

```
char a[10]  
a[0] =
```

```
for  
if (char a[i] == char a[j])  
or a[j] == char a[i]  
a++;
```

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FAST - National University of Computer and Emerging Sciences, Lahore Campus

**Midterm 2: CS 116 Introduction To Computing**

Time: 60 minutes, Max. Marks: 50, Spring 2016

**: Programming for Engineers I**

[Marks: 20]

**Q. #1:**

Euclid's method for finding the greatest common divisor (GCD) of two positive integers consists of the following steps:

- Step 1: Divide the larger number by the smaller and retain the remainder.
- Step 2: Divide the smaller number by the remainder, again retaining the remainder.
- Step 3: Continue dividing the previous remainder by the current remainder until the remainder is zero, at which point the last non-zero remainder is the GCD.

For example, if the two positive integers are 84 and 49, you have the following:

- Step 1:  $84/49$  yields a remainder of 35.
- Step 2:  $49/35$  yields a remainder of 14.
- Step 3:  $35/14$  yields a remainder of 7.
- Step 3:  $14/7$  yields a remainder of 0.

Therefore, the last non-zero remainder, which is 7, is the GCD of 84 and 49.

Write a C++ function GCD [with three input parameters and a return type of void] that calculates GCD of two integers and use them in the main program.

[Marks: 30]

**Q. #2:**

Write C++ function(s) that performs the following tasks:

- a) Takes array and its size as input argument and print all odd numbers and total count of odd numbers in an array. The function must notify the user in case there is no odd number.
- b) Takes two arguments: firstNum and secondNum (firstNum must be less than secondNum). Print all odd numbers between firstNum and secondNum.
- c) Takes an integer and checks whether it is prime or not. The function must notify the user accordingly. [A prime number is a natural number greater than 1 that has no positive divisors other than 1 and itself.]

Write a sample main to test these functions.

(49)

(84)